

UNIVERSITY OF ELDORET
OFFICE OF THE DEPUTY VICE CHANCELLOR (ASA)
SCHOOL OF SCIENCE
DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

COURSE CODE: MATHS 313...PRINCIPLES OF OPERATION RESEARCH

CAT 1 12/11/2024 INSTRUCTIONS: ANSWER ALL QUESTIONS

1. Define Operation Research and explain two features of Operation Research [3Mks]
2. Highlight and explain three components of linear programming model. [3Mks]
3. Write its dual model and solve its primal using Simplex Method [6Mks]

$$\text{Minimize } Z = 2y_1 - 3y_2 + 6y_3$$

$$\text{Subject to : } 3y_1 - y_2 + 2y_3 \geq 7$$

$$2y_1 + 4y_2 \leq 8$$

$$-4y_1 + 3y_2 + 8y_3 \geq 10$$

$$y_1, y_2, y_3 \geq 0$$

4. A factory makes two types of components for the motor industry. Two machines are required for the manufacture of each component. Component A requires one and a half hours on machine 1 and one hour on machine 2. Component B requires two hours of machine 1 and 3 hours of machine 2. In a given day, there are at least 9 hours and at most 12 hours available for machines 1 and 2 respectively. In addition, the material required for component A is in short supply and only sufficient material for 6 units in the period is available. If the profit contribution from component A is \$20 and component B is \$30.

- (i) Develop a linear programming model to this problem [3 marks]
 - (ii) Determine the optimum production schedule graphically [3 marks]
 - (iii) Determine the contribution for the period graphically [2 marks]
5. A furniture company produces a variety of products. One department specializes in wood tables, chairs and book cases that are made using three resources namely;

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labour, wood and machine time. The department has 60 hours of labour available each day, 16 hours of machine time and 400 board feet of wood. A consultant has developed a LP model for the department as follows:

Let X_1, X_2, X_3 represent the quantity of tables, chairs and book cases,

$$\text{Maximize } Z = 40X_1 + 30X_2 + 45X_3$$

$$\text{Subject to: } 2X_1 + X_2 + 2.5X_3 \leq 50 \dots \text{Labour}$$

$$0.8X_1 + 0.6X_2 + 1.0X_3 \leq 16 \dots \text{Machine Time}$$

$$30X_1 + 20X_2 + 30X_3 \leq 400 \dots \text{Wood}$$

$$X_1 \geq 0, X_2 \geq 0, X_3 \geq 0$$

Answer the following questions posed by the departmental manager

- i. What is the purpose of the model? [1Mk]
- ii. What does the number 40 in the objective function stand for? [1Mk]
- iii. What does the term $40X_1$ in the objective function stand for? [2Mk]
- iv. What does the number 30 in the third constraint for? [2Mk]
- v. What does the term $30X_3$ in the objective function stand for? [2Mk]
- vi. Explain the significance of the constraint $X_1 \geq 0, X_2 \geq 0, X_3 \geq 0$ [2Mk]